

Name: Fenni Jambeko Shatsilika
Student number: 226176614
Course: Computational Game Theory

Assignment: Paper Review

Paper: Bowling, M., Burch, N., Johanson, M., and Tammelin, O. (2017). *Heads-Up Limit Hold'em Poker Is Solved*. Communications of the ACM, 60(11): 81–88.

1. Summary of the Paper

1.1 The problem being addressed

The paper tackles the problem of solving an imperfect-information game that is genuinely played competitively by humans. "Solving" here means computing a game-theoretically optimal strategy, one that cannot lose against any opponent in a fair match rather than merely beating strong human players (as Deep Blue or Watson did in their domains).

Before this work, every non-trivial game played competitively by humans that had been solved was a perfect-information game (e.g., Connect Four and checkers), in which all players observe everything that has happened. Imperfect-information games are substantially harder: players act on partial knowledge (e.g., not seeing the opponent's private cards), so the standard game-tree search techniques used for perfect-information games do not apply.

The specific target is Heads-Up Limit Texas Hold'em (HULHE); the two-player, fixed-bet, capped-raise variant, and the smallest version of poker played for money. Despite being "smallest," it is enormous: roughly 3.16×10^{17} reachable game states and 3.19×10^{14} decision points (information sets), about three orders of magnitude larger than any imperfect-information game solved before it. The two central obstacles are therefore memory (storing strategy and regret values naively would require about 262 TB) and computation (the scale-up implies a large increase in required compute).

1.2 Main contribution of the work

The headline contribution is the announcement that HULHE is essentially weakly solved; the first non-trivial imperfect-information game played competitively by humans to be solved. The solving program is named Cepheus.

Because the game-theoretic value of an imperfect-information game is real-valued and only realized in expectation, the authors define a pragmatic standard: a game is "essentially solved" if the computed strategy's residual exploitability is small enough that a *human lifetime* of play (modelled as 200 hands/hour, 12 hours/day for 70 years) could not statistically distinguish it from a perfect solution at 95% confidence. This translates to a threshold of 1 milli-big-blind per game (mbb/g).

The technical enabler is a new algorithm, CFR⁺, a variant of Counterfactual Regret Minimization (CFR). CFR⁺ differs from standard CFR in three ways:

- It uses regret-matching⁺, in which accumulated regrets are clamped to be non-negative, so an action that has looked poor can be re-selected immediately once it proves useful again, rather than waiting many iterations.

- It performs exhaustive, full-tree iterations instead of the Monte-Carlo sampling typically used by CFR variants.
- It empirically converges using the players' current strategies, so the algorithm can skip the usual step of computing and storing the time-averaged strategy.

To overcome the memory wall, the authors store strategies and regrets with fixed-point compression (scaling to integers and ordering them for compressibility, achieving roughly 13:1), reducing storage from ~262 TB to under 11 TB held on local disk, with the game partitioned into subgames streamed from disk.

1.3 Experimental / theoretical results

- **Solution quality.** Run on a cluster of 200 nodes (24 cores each), the computation took 1,579 iterations over 68.5 days (~900 core-years) across 110,565 subgames. The resulting strategy reached an exploitability of 0.986 mbb/g, below the 1 mbb/g threshold, hence "essentially weakly solved."
- **Dealer advantage.** Using per-position exploitability, the authors derive exact bounds on the game value: the dealer wins by between 87.7 and 89.7 mbb/g, formally confirming the long-held belief that the dealer has a significant advantage.
- **Strategic insights.** The near-optimal strategy almost never "limps" (open-calls), validating conventional wisdom, but also rarely "caps" early betting as the dealer and plays a wider non-dealer range than typical human players, suggesting areas where human play could improve.
- **In-game performance (CACM extension).** Against a series of historical computer agents (PsOpti4 through CFR-BR), Cepheus illustrates that low in-game loss and low worst-case exploitability are *not* the same: agents with very different exploitabilities lost to Cepheus by similar small margins. Against humans, across ~39,564 usernames and ~3.56 million games, Cepheus won at 169.9 ± 5.2 mbb/g overall; even against the strongest tier of players (after filtering for self-selection and variance) it won about 87 mbb/g.

2. Related Work

The paper places itself within a long line of work on solving games, which can be grouped as follows.

Perfect-information milestones and AI game-playing. The authors trace games as a testbed for AI from Babbage, Turing, and Shannon's early chess work, through the marquee achievements of beating humans, Chinook in checkers (Schaeffer et al.), Deep Blue defeating Kasparov in chess (Campbell et al.), and Watson on *Jeopardy!* (Ferrucci). They distinguish these from *solving* a game, citing the solving of Connect Four (Allis) and checkers (Schaeffer et al., 2007). Allis's three-level taxonomy, ultra-weakly, weakly, and strongly solved, frames their own claim.

Game-theoretic foundations. Poker motivated foundational game theory: Borel and von Neumann's work on bluffing, von Neumann and Morgenstern's *Theory of Games and*

Economic Behavior, Nash equilibrium as the central solution concept, Zermelo's 1913 solvability result, and the minimax theorem.

Linear-programming approaches. Early methods converted the extensive-form game to normal form and solved an LP, but the number of deterministic strategies is exponential in the number of information sets, making this intractable beyond toy games such as Kuhn poker. The sequence-form representation (Romanovskii; Koller and Megiddo; Koller, Megiddo, and von Stengel) was a major advance, allowing an extensive-form game to be solved as an LP in time polynomial in the size of the game representation. Billings et al. (2003) applied this to simplified HULHE for the first competitive poker program, and Gilpin and Sandholm solved Rhode Island Hold'em using sequence-form LP plus automated abstraction.

Counterfactual Regret Minimization (CFR). Introduced by Zinkevich et al. (2007/2008), CFR minimizes regret per information set rather than over the exponential strategy space, making memory only *linear* in the number of information sets, a step-change comparable to the sequence-form advance. CFR became the dominant approach in the Annual Computer Poker Competition (begun 2006). Subsequent refinements include Monte-Carlo CFR / sampling variants (Lanctot et al.; Johanson et al., 2012) and large-scale commodity-hardware implementations such as Slumbot (Jackson, 2012, reaching $\sim 3.8 \times 10^{10}$ information sets). An alternative line, Nesterov's excessive gap technique applied to extensive-form games (Gilpin, Hoda, Peña, and Sandholm), offered better asymptotic complexity but less flexibility; Waugh and Bagnell later argued the two families are closely related.

Measuring and improving exploitability. Johanson et al. (2011) made it feasible to compute the exact worst-case exploitability of large-game strategies, and CFR-BR (Johanson et al., 2012) explicitly solved for the abstract strategy with the lowest real-game exploitability, the closest equilibrium approximation prior to this work.

Direct antecedent. The algorithm itself, CFR⁺, was proposed by Tammelin (2014), with its theoretical guarantees (Theorem 1 and regret-matching⁺ analysis) established in the follow-up by Tammelin, Burch, Johanson, and Bowling (2015).

(Within the course reading list, Tammelin (2014) [paper #4] is the immediate algorithmic foundation of this work, and Lanctot et al. (2017) [paper #3] continues the same CFR/multi-agent learning research program.)

3. Limitations and Critique

Despite the results of this paper, several limitations qualify its scope and its broader claims.

1. The "essentially weakly solved" claim is approximate, and the threshold is a convention. The final strategy has a residual exploitability of 0.986 mbb/g; close to, but not exactly, zero. The game is solved only in the sense of being statistically indistinguishable from optimal *over a human lifetime at 95% confidence*. A non-human adversary playing far more hands, or a future best-response computation, could in principle establish that the strategy is genuinely sub-optimal. The 1 mbb/g bar is reasonable but ultimately a chosen statistical convention rather than an exact solution.

2. Narrow scope. The result covers only HULHE; the *smallest* poker variant played for money. The much larger no-limit and multiplayer variants remain unsolved and are many orders of

magnitude bigger. The paper frames CFR⁺ as a general advance for "game-theoretic reasoning in large-scale models," but the demonstration is confined to a single two-player game.

3. Dependence on the two-player zero-sum structure. The strong properties exploited here, a unique game value, interchangeable equilibria, and the meaningfulness of "unexploitable" play, hold specifically for two-player zero-sum games. In general-sum or multiplayer settings a Nash equilibrium is neither unique nor necessarily a good strategy, and CFR's convergence guarantees do not transfer. The "solving" achievement is therefore tied to a special game class.

4. An unexploitable strategy is not a maximally winning one. A Nash-equilibrium strategy is designed to be *incapable of losing*, not to *maximize winnings against weak opponents*. It does not adapt to exploit opponents' mistakes. The in-game results show Cepheus winning convincingly, but defeating weak opponents is not what the equilibrium optimizes for, so those results do not strengthen the solving claim itself.

5. Heavy computational cost and limited reproducibility. The solution required ~900 core-years on a 200-node cluster with ~11 TB of disk and game-specific compression (13:1) and subgame decomposition. CFR⁺ performs exhaustive full-tree traversals rather than sidestepping the scaling problem; extending the approach to larger games would still require abstraction, which reintroduces approximation error that is *not* bounded by the guarantees demonstrated here. The result is essentially out of reach for independent replication without comparable resources.

6. Theory lagged practice at announcement. A key efficiency choice, using the players' current strategies and skipping the average strategy, was justified empirically in the original work; the formal support for CFR⁺ and regret-matching⁺ (the paper's Theorem 1) appeared only in a subsequent publication. At the time of the original solving announcement, part of the method rested on observed behaviour rather than proof.

7. The human evaluation is observational, not controlled. The CACM in-game section is informative but weak as evidence: the authors themselves note self-selection (most players never finished a 100-game match), the bias from players quitting while ahead, unequal sample sizes, and high variance. The tiered analysis mitigates these issues but cannot fully remove them, so the human results are best read as illustrative rather than rigorous.

Overall assessment. Within its stated scope, the paper is rigorous, honest about its assumptions, and clearly a milestone, the first competitively played imperfect-information game to be solved, and a genuine algorithmic advance in CFR⁺. Its main weaknesses are not errors but boundaries: the approximate nature of "essentially solved," the narrowness of the two-player zero-sum HULHE setting, the very large compute cost, and the gap between solving a game and playing it optimally against real, fallible opponents.

References

- Billings, D., Burch, N., Davidson, A., Holte, R. C., Schaeffer, J., Schauenberg, T., & Szafron, D. (2003). Approximating game-theoretic optimal strategies for full-scale poker. *IJCAI*, 661–668.
- Bowling, M., Burch, N., Johanson, M., & Tammelin, O. (2015). Heads-up limit hold'em poker is solved. *Science*, 347(6218), 145–149.
- Bowling, M., Burch, N., Johanson, M., & Tammelin, O. (2017). Heads-up limit hold'em poker is solved. *Communications of the ACM*, 60(11), 81–88. (paper under review)
- Campbell, M., Hoane, A. J., & Hsu, F. (2002). Deep Blue. *Artificial Intelligence*, 134, 57–83.
- Ferrucci, D. (2012). Introduction to "This is Watson." *IBM Journal of Research and Development*, 56(3.4), 1:1–1:15.
- Gilpin, A., Hoda, S., Peña, J., & Sandholm, T. (2007). Gradient-based algorithms for finding Nash equilibria in extensive form games. *WINE*, 57–69.
- Gilpin, A., & Sandholm, T. (2007). Lossless abstraction of imperfect information games. *Journal of the ACM*, 54(5).
- Jackson, E. (2012). Slumbot: An implementation of counterfactual regret minimization on commodity hardware. *Computer Poker Symposium*.
- Johanson, M., Bard, N., Burch, N., & Bowling, M. (2012). Finding optimal abstract strategies in extensive form games. *AAAI*, 1371–1379.
- Johanson, M., Bard, N., Lanctot, M., Gibson, R., & Bowling, M. (2012). Efficient Nash equilibrium approximation through Monte Carlo counterfactual regret minimization. *AAMAS*.
- Johanson, M., Waugh, K., Bowling, M., & Zinkevich, M. (2011). Accelerating best response calculation in large extensive games. *IJCAI*, 258–265.
- Koller, D., & Megiddo, N. (1992). The complexity of two-person zero-sum games in extensive form. *Games and Economic Behavior*, 4(4), 528–552.
- Koller, D., Megiddo, N., & von Stengel, B. (1996). Efficient computation of equilibria for extensive two-person games. *Games and Economic Behavior*, 14(2).
- Koller, D., & Pfeffer, A. (1997). Representations and solutions for game-theoretic problems. *Artificial Intelligence*, 94, 167–215.
- Kuhn, H. W. (1950). Simplified two-person poker. In *Contributions to the Theory of Games* (Vol. 1, pp. 97–103). Princeton University Press.
- Lanctot, M., Zambaldi, V., Gruslys, A., Lazaridou, A., Tuyls, K., Pérolat, J., Silver, D., & Graepel, T. (2017). A unified game-theoretic approach to multiagent reinforcement learning. *NIPS'17*, 4193–4206.
- Nesterov, Y. (2005). Excessive gap technique in nonsmooth convex minimization. *SIAM Journal on Optimization*, 16(1), 233–249.

- Romanovskii, I. V. (1962). Reduction of a game with complete memory to a matrix game. *Soviet Mathematics*, 3, 678–681.
- Schaeffer, J., Burch, N., Björnsson, Y., Kishimoto, A., Müller, M., Lake, R., Lu, P., & Sutphen, S. (2007). Checkers is solved. *Science*, 317(5844), 1518–1522.
- Tammelin, O. (2014). Solving large imperfect information games using CFR+. *arXiv:1407.5042*.
- Tammelin, O., Burch, N., Johanson, M., & Bowling, M. (2015). Solving heads-up limit Texas hold'em. *IJCAI*, 645–652.
- von Neumann, J., & Morgenstern, O. (1947). *Theory of Games and Economic Behavior* (2nd ed.). Princeton University Press.
- Waugh, K., & Bagnell, J. A. (2015). A unified view of large-scale zero-sum equilibrium computation. *AAAI Workshop on Computer Poker and Imperfect Information*.
- Zermelo, E. (1913). Über eine Anwendung der Mengenlehre auf die Theorie des Schachspiels. In *Proceedings of the Fifth International Congress of Mathematicians* (pp. 501–504).
- Zinkevich, M., Johanson, M., Bowling, M., & Piccione, C. (2008). Regret minimization in games with incomplete information. *NIPS*, 905–912.